

Battery Simulator – Project Report

The goal of the project was to design and construct a circuit that allows for the simulation of the battery charging process and testing its reaction to a variable load (constant current constant voltage).

The circuit consists of two sections – a voltage section allowing for free adjustment of the input voltage in the range from 2.9 to 4.2 volts, and a current section designed to maintain and regulate the load current of the circuit up to 1 ampere.

Components used in the circuit:

- LDO voltage regulator LT3080 1.1A
- LDO voltage stabilizer MIC29302
- Resistor R1 – 0.68 Ω
- Resistor R2 – 1 Ω
- Resistor R6 – 200 Ω
- Resistor R7 – 82 Ω
- Resistor R33 – 220 k Ω
- Resistor R88 – 220 Ω
- Potentiometer R3 – 100 k Ω
- Potentiometer R8 – 100 Ω
- Capacitor C1 – 4.7 μ F
- Capacitor C2, C3 – 1 μ F
- Capacitor C6 – 22 μ F
- Capacitor C7 – 10 μ F
- Schottky diode 1N5822

Description of circuit operation:

The voltage from the power supply goes to the input of the MIC29302 stabilizer. Resistors R6 and R7 form a voltage divider setting the voltage value at the stabilizer output to 4.2 volts. Using potentiometer R8, connected in series with R7, this voltage can be adjusted to a minimum value of 2.9 volts. Capacitors C6 and C7 are designed to stabilize the input and output of the integrated circuit and eliminate noise, while the diode protects it against reverse current. When connecting a load to the circuit at the X2 connector, a constant load current value is ensured by the LT3080 regulator circuit. Using potentiometer R3 in the set branch, the reference voltage is set to a maximum value of about 0.7V, which in combination with resistors R1 and R2 at the output provides an adjustable stable load current. Resistors R33 and R88 are connected in parallel to potentiometers R3 and R8 to adjust the resistance range of the potentiometers to the desired value. Capacitors C2 and C3 filter noise from the voltage stabilizer output side and the load. Capacitor C1 stabilizes the regulator output.

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